Johannes Schwab

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Google Scholar

Research interests

Machine learning applied to finance, with a focus on asset pricing: high-dimensional tests of asset pricing models, the generalization of machine-learning models, and language models free of lookahead bias

Education

École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland
September 2023 – present
PhD in Finance
Advisor: Prof. Semyon Malamud (Swiss Finance Institute, EPFL)

École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland
September 2019 – August 2021
Master of Science in Financial Engineering, GPA: 5.8/6
Master's thesis: Use of machine learning techniques for portfolio insurance (at Vaudoise Assurances)

University of Lausanne (UNIL), Faculty of Business and Economics (HEC), Lausanne, Switzerland
September 2016 – June 2019
Bachelor of Science (BSc) in Economics, GPA: 5.6/6

Publications

Scaling Point-in-Time Language Models **NeurIPS 2026**
with Bryan Kelly, Semyon Malamud and Teng Andrea Xu arXiv:2607.11889 • NBER WP 35247

- Shows that training decoder-only transformers up to 4B parameters on 1T chronologically filtered tokens largely closes the performance gap between lookahead-free language models and their unconstrained counterparts
  Presented at: EPFL Computational Future seminar 2026 (F. Kaplan)
  To be presented at: NeurIPS 2026 (poster)

Working papers

Training NTK to Generalize with KARE *2025*
with Bryan Kelly, Semyon Malamud and Teng Andrea Xu arXiv:2505.11347 • SFI Research Paper 25-51

- Proposes optimizing the neural tangent kernel explicitly rather than implicitly through gradient descent, yielding kernels that match or outperform the deep networks they are derived from

A Test of the Efficiency of a Given Portfolio in High Dimensions *2025*
with Mikhail Chernov, Bryan Kelly and Semyon Malamud NBER WP 33565 • SFI Research Paper 25-26

- Extends the Gibbons–Ross–Shanken test to settings where test assets far outnumber the sample, using Random Matrix Theory to correct the bias induced by regularized covariance estimation
  Presented at: SoFiE 2025 (Paris), ESSFM 2025 (Gerzensee)

Teaching

Teaching Assistant, EPFL, Master in Financial Engineering
September 2023 – present

- Stochastic Calculus
- Machine Learning in Finance

Honors and awards

- **Best Teaching Assistant, 2026 —** awarded by student vote in the EPFL Master in Financial Engineering, for Stochastic Calculus and Machine Learning in Finance
- **Swiss Life Award, 2021 —** awarded to the student graduating with the highest GPA in the EPFL Master of Science in Financial Engineering

Industry experience

Quantitative Analyst, Taranis – Hedge Fund, Geneva, Switzerland
March 2022 – May 2023

- Designed a backtesting environment for option-based strategies on a SQL database of over 1 billion data points, developed a Python library for the statistical analysis of trading signals, and co-designed an automated trading bot

Asset Management Intern, Vaudoise Assurances, Lausanne, Switzerland
March 2021 – December 2021

- Calibrated an option-based portfolio insurance (OBPI) strategy and built Python tools for option pricing, backtesting and reporting

Skills

- **Programming** — Python (PyTorch, NumPy, pandas, scikit-learn), MATLAB, SQL, LaTeX
- **High-performance computing** — Multi-node GPU training (128 GPUs at CSCS), MPI, NCCL, SLURM

Languages

- **French** — Native
- **English** — Fluent (C1)
- **German** — Upper intermediate (B2)

Last updated: October 20, 2026